Morphometry of the Thoracic and Lumbar Spine Related to Transpedicular Screw Placement for Surgical Spinal Fixation

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Vertebral transpedicular screws provide secure attachment for posterior spinal fixation devices. Screw design details, biomechanics, and implantation safety depend upon anatomic constraints, especially from the pedicle and body. Previous morphometric data were limited; thus, a retrospective study was undertaken using computerized axial tomograms (CT) of 91 vertebrae (T9–L5). In addition, eight cadaver vertebrae were CT scanned and then cut transversely to compare x-ray measurements with direct physical measurements. Measured parameters included pedicle width, pedicle length, angle of pedicle axis to sagittal plane, and transpedicular cortex-to-cortex chord length. Good correlation is shown to occur between CT scan and direct physical measurements of human vertebrae. Implications for spinal implant screw dimensions and safety of implantation are discussed. Comparison with previously available data is made. [Key words: spinal fixation, pedicle screws, morphometry, CT scanning]

CURRENTLY AVAILABLE SURGICAL implants for management of single- or few-motion segment problems (eg, fracture, degeneration, spondylolisthesis, tumor) have certain important shortcomings. The single biggest problem is the practice of spanning of five, six, or even seven vertebrae when as few as two vertebrae are actually involved in the pathology. Harrington distraction instrumentation, originally devised for multimotion-segment problems (eg, scoliosis), is by far the most commonly used implant. Because the distraction hooks are not rigidly attached to a vertebra the system depends on intact soft tissues to maintain the necessary tensile forces to prevent hook dislodgement, and the hooks must be widely separated (up to seven vertebrae spanned) to provide adequate protection at the site of instability. Wiring together of lamina and rod (either Harrington or Luque) at each spanned vertebra still requires spanning an unnecessarily long spinal segment, again because there is not a three-dimensionally rigid link between rod and vertebra. Roy-Camille plates[6-8] represent a major improvement but also have the same drawback of an excessively long spanned length. The screws are not rigidly linked to the plate but depend on compressive forces between plate and subjacent irregular bony surface to prevent "toggle" of the screws in the plate holes. The Magerl external spinal fixator[4,5] overcomes this problem but does have the attendant difficulties of device prominence and risk of infection. The Dunn anterior fixator[1] is limited in its application by the need to use an anterior surgical approach and its potentially close proximity to the aorta. A number of implants have been devised or are being developed and have been reviewed extensively elsewhere.[3]

Thus, a design effort was undertaken in 1981 to overcome these shortcomings.[3] To minimize the length of spinal segment involved, a three-dimensionally rigid grip on each vertebra is needed. Placement of a screw through the pedicle into vertebral body appears to be a very good way to accomplish this. For this reason, the details of pedicle morphometry become important in defining anatomic design constraints. Some data exist on this subject,[9] but these are limited in certain ways: no standard deviations or ranges are reported, and pedicle dimensions were given only in the sagittal and coronal planes even though the pedicles are often oriented in an oblique direction and screw placement may be desired along this oblique direction. Because of these and other limitations to available data, the current study was undertaken to define certain important morphometric details relative to the human vertebral pedicle. Since this study was completed, additional data have been published on certain aspects of pedicle anatomy.[10]

MATERIALS AND METHODS

A retrospective review was made of all computerized tomographic (CT) scans of the thoracic and lumbar spine performed at the Medical Center Hospital of Vermont between September 1982 and November 1983. Data were recalled from magnetic storage diskettes, and the transverse section on which the left and right pedicle appeared largest was selected for printing on x-ray film at 80% lifesize. This image was referred to as the midpedicle cut. Excluded were vertebrae that were fractured or adjacent to the internal fixation devices that had been placed (metal-generated artifact made measurements less reliable). All scans had been obtained using a Siemens Somatom 2 third-generation CT scanner, using 4-mm cuts and a 5-second scan time.

Ninety-one vertebrae from 41 patients were selected for analysis. The patient age range was from 18 to 75 years, (average age: 36 years, median age: 28 years). Fourteen patients were women and 27 were men. The final diagnoses were as follows: ten fractured vertebrae, seven herniated discs, five bulging discs, three unspecified cord lesions, two bony degenerative changes, and fourteen normal studies. Of the normal studies, six were performed for suspected herniated disc, five on multitrauma patients, two for suspected fracture, and one for suspected bony degenerative change.

The 80% lifesize CT images were backlighted with a horizontally mounted viewbox, and a 0.5-mm soft lead mechanical pencil was used to draw reference lines directly on the x-ray film. The anteroposterior (AP) midline axis was defined by visual inspection with the assistance of a grid-patterned overlay (1-mm × 1-mm graph velum), the orientation of which was adjusted as closely as possible so that a single line fell transversely along both transverse processes

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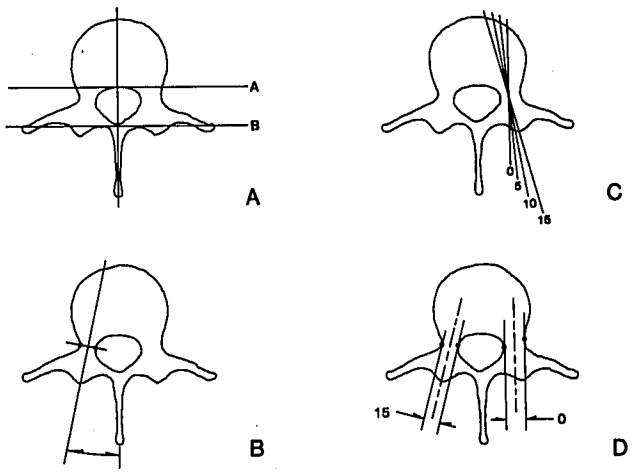


Fig 1. Construction lines used to obtain measurements from vertebral CT scans. **A,** Vertebral body length from anterior cortex to line A; pedicle length from line A to posterior cortex or to line B (facet corrected). **B,** Pedicle axis angle measured from sagittal plane (anteromedial positive); pedicle diameter along a perpendicular to axis. **C,** Screw-path length (or chord length) from anterior to posterior cortex at 0°, 5°, 10°, or 15° posterolaterally from the sagittal plane. **D,** Pedicle diameter at 0° and 15°: note that bone contact points do not fall along a common perpendicular to the axis.

while simultaneously a line perpendicular to this bisected the area of both the vertebral body and the spinal canal (Figure 1A). Reference line A (Figure 1A) was drawn perpendicular to the AP midline axis and tangent to the most anterior projection of the spinal canal. Line B (Figure 1A) was drawn parallel to line A and along the flattest portion of the posterior cortex of the lamina. The area between lines A and B was defined as the pedicle, the area anterior to line A was defined as the body, and the area posterior to line B in the region of the pedicle represents the facets. The pedicle longitudinal axis (Figure 1B) was drawn through the middle of the pedicle by visual best fit and with the aid of the grid overlay. At the narrowest width of the pedicle, a line was drawn perpendicular to the longitudinal axis. Through this intersection point were drawn the four angular transpedicular chord lengths at 0°, 5°, 10°, and 15° (defined below and shown in Figure 1C).

Linear measurements were taken from the CT images using dividers and a 0.5-mm ruled scale. All measurements were read to the nearest 0.33 mm and corrected by a factor of 1.25 to determine lifesize dimension. Angular measurements were made to the nearest 0.5° using a hand-held protractor. Angulation anteromedially was defined as positive; angulation anterolaterally was defined as negative.

Chord lengths (Figure 1C) were measured from reference line B along the 0°, 5°, 10°, and 15° axes to a point intersecting the anterior cortex. The reason for using reference line B instead of the posterior cortex itself was to correct for variability in facet anatomy as it appears on CT scan.

Pedicle length was measured along the 0°, 5°, 10°, and 15° axes. Only the 5° data are presented here, because the other values were so similar. This measurement was obtained from the posterior cortex to line A.

Pedicle diameter (Figure 1, B and D) was measured in three ways: perpendicular to the 0° axis, the pedicle axis, and the 15° axis. For the second of these methods, the diameter was measured along a single perpendicular to the pedicle axis (Figure 1B). For the 0° and 15° axis (Figure 1D), the diameter was measured as the distance between two lines drawn parallel to either the 0° or 15° axes, respectively, and tangent to the medial and lateral pedicle cortices. Note that the two points of tangency do not, in general, fall along a single perpendicular to the measurement axis. This method was used for the off-pedicle-axis measurements to reflect the true diameter of a screw that could be placed through the pedicle into the vertebral body without cortical disruption.

Vertebral body length was measured from line A to the anterior cortex, along the AP midline axis. This dimension is readily measurable on lateral view roentgenograms as well as on CT scans. Correlation of vertebral body depth to chord length was performed to allow the latter (important in choosing length of screw to be implanted) to be predicted from lateral view roentgenograms, in case CT scans are unavailable.

To demonstrate the accuracy of the CT measurements, cadaver vertebrae were measured both by direct measurements as well as by CT scannings. Eight cadaver vertebrae (1 twelfth thoracic, 3 second lumbar, 3 third lumbar, and 1 fourth lumbar) had their inferior end-plates embedded in methylmethacrylate and mounted in jigs to allow positioning for CT scanning. CT scans were performed in an identical manner to those used in the earlier patient studies. Data were collected on these CT images in the same fashion as previously described. The eight vertebrae were then sectioned through the midpedicle in the transverse plane (parallel to the end-plates) using a band saw. Measurements of chord length along the 0° axis, pedicle diameter (perpendicular to pedicle axis), and body length were compared with the CT data.

The factors of patient age and bone density were not specifically analyzed here. Bone density determinations were not available

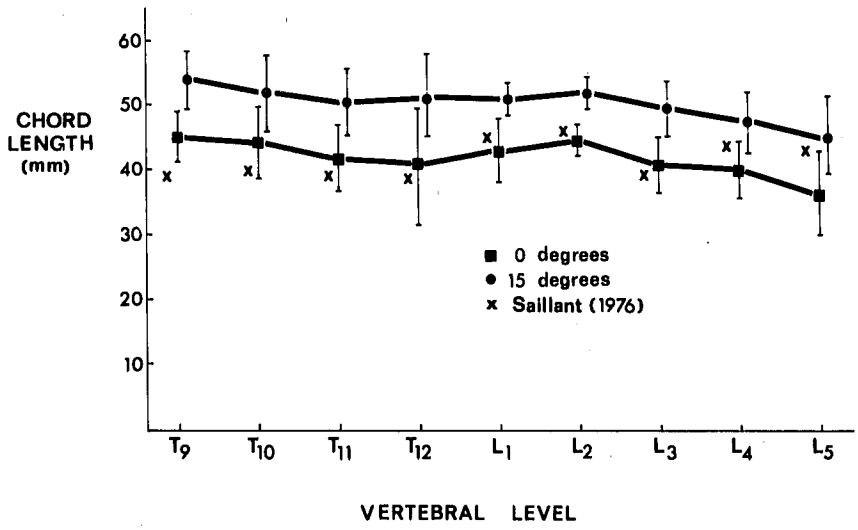


Fig 2. Screw-path length (chord length) measured along a line at 0° and 15° relative to the sagittal plane (see Figure 1C), and compared with means ("x") from Saillant (1976).

Table 1. Chord Length Distribution for Each Vertebral Level and for 0° and 15° Axes

Level	# of Pedicle	Axis angle	% of Pedicles with specified range in chord lengths									
			20-25 mm	25-30 mm	30-35 mm	35-40 mm	40-45 mm	45-50 mm	50-55 mm	55-60 mm	60-65 mm	65-70 mm
T9	14	0°				7	43	36	14			
		15°						21	36	29	14	
T10	18	0°				22	39	17	22			
		15°					11	44	22	11	11	
T11	22	0°			18	23	32	18	9			
		15°					18	41	23	14	4	
T12	24	0°	12	5	8	12	29	25				
		15°					17	33	21	29		8
L1	22	0°		4	4	9	54	27				
		15°						41	50	9		
L2	14	0°				7	43	50				
		15°						14	86			
L3	24	0°			5	54	21	17	5			
		15°					17	38	38	8		
L4	24	0°		5	8	42	29	17				
		15°					25	46	21	8		
L5	20	0°		15	50	5	15	15				
		15°				25	30	15	25	5		
Mean %		0	1.3	3.2	10.3	20.1	33.8	24.7	5.6			
		15				2.8	13.1	32.6	35.8	12.6	3.2	0.9
		both	0.6	1.6	5.2	11.4	23.4	28.6	20.7	6.3	1.6	0.4

retrospectively, and the choice of subjects was not statistically balanced with respect to age.

RESULTS

Chord length is shown in Figure 2 for measurement along the 0° and 15° axes. Means and standard deviation (SD) error bars are shown for each vertebral level. These are compared with the data of Saillant[9], which consisted only of mean values. Close agreement between our means and those of Saillant can be seen. Note that chord length is fairly constant from T9 to L5 for both the 0° and 15° lengths. Mean values over all vertebral levels are 42 mm at 0° and 50 mm at 15°. The standard deviations are all small. The distribution of chord lengths is shown in Table 1 for measurements along both the 0° and 15° axes. Note that the chord lengths tend to be tightly clustered. For the 0°axis, 78.6% of the lengths fall between 35.1 and 50 mm, and for the 15° axis, 68.4% fall between 45.1 and 55 mm. For both axes combined, 84.1% of the lengths fall between 35.1 and 55 mm. Table 1 also shows the generally shorter chord length for the 0° axis than for the 15° axis for each vertebral level.

Pedicle length, shown in Figure 3, ranges between approximately 15 mm and 25 mm, is nearly constant between T9 (mean: 20.18 mm) and T12 (mean: 18.06 mm), then gradually decreases to L5 (mean: 15.00 mm). This finding is consistent with the decreasing volume of the cauda equina in the spinal canal as one progresses caudally. Expressed as a percentage of the chord length along the 0° axes, the pedicle lengths range from a maximum of 46% (T12 level) to a minimum of 40% (T10 level).

Vertebral body length (Figure 4) is seen to be almost constant, with a mean overall vertebral level from T9 to L5 of 32 mm. This measurement was obtained primarily to allow prediction of chord length (needed to estimate screw length for surgical implant placement), even if only lateral view roentgenograms are available. The factor by which to multiply the vertebral body length to obtain the chord length for the corresponding approach angles is as follows (mean ± SD): for 0°, 1.32 ± 0.12; for 5°, 1.44 ± 0.09; for 10°, 1.53 ± 0.09, and for 15°, 1.60 ± 0.09. The overall mean factor value across all approach angles is 1.50.

Pedicle diameter (Figure 5) is seen to be fairly constant between T9 and L1, with a value of approximately 7 mm. A gradual increase then occurs from L1 and L5 where it is approximately 13 mm. The

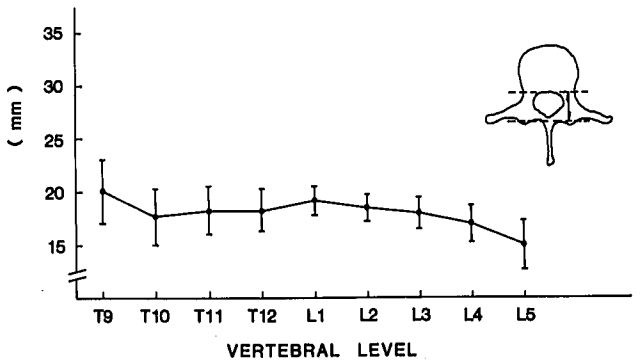


Fig 3. Pedicle root length, facet corrected. Pedicle length means ± 1 SD measured along a 0° axis, between lines A and B (see Figure 1A).

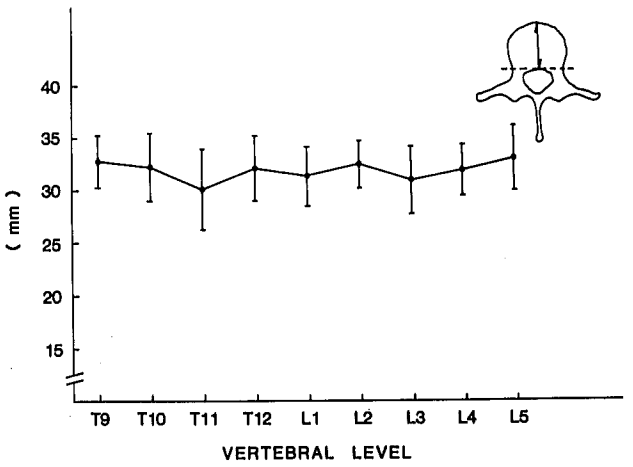


Fig 4. Vertebral body length means ± 1 SD measured along a 0° axis between line A and anterior (see Figure 1A).

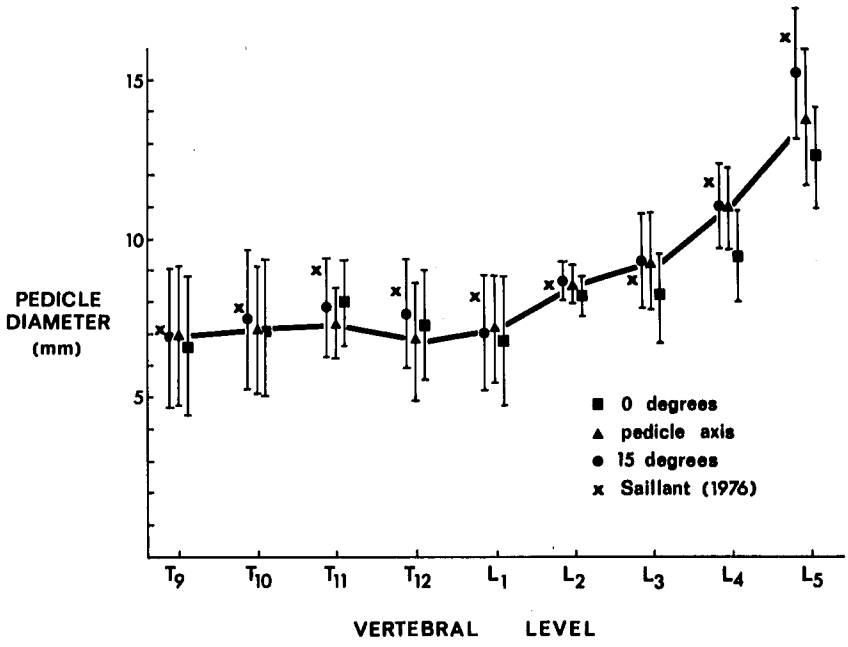


Fig 5. Pedicle diameter means ± 1 SD for each vertebral level, measured three different ways (see Figure 1, B and D) and compared with means ("x") from Saillant (1976).

three different methods of measurement (perpendicular to the 0° axis, the pedicle axis, and the 15° axis) give quite similar results, except at L4 and L5 where some divergence does occur. Even at L5, however, the orientation giving the smallest value (0° axis) still yields a diameter that is quite large (12.6 mm). The mean values reported by Saillant[9] are in close agreement, although they tend to be somewhat higher. Table 2 shows the distribution frequency of pedicle diameter. It can be seen that a 5-mm diameter screw will fit through all but the few very smallest pedicles throughout T9 through L5. Screws of 6 mm, 7 mm, or even larger will fit through most of the lower lumbar pedicles.

Pedicle angle (Figure 6) follows a pattern fairly similar to that of pedicle diameter: fairly constant throughout T9 to L1, then gradually increasing from L1 to L5. The pedicle angle is approximately 0-10° for T9-L1, 10-20° for L1-L4, and 20-30° for L4-L5. The standard deviations are fairly small throughout the vertebral levels studied. Significant differences are seen relative to the mean values reported by Saillant.[9]

Confirmation of the accuracy of measurements obtained from CT scans is shown in Figure 7. Here the values obtained by direct caliper measurement are plotted against those obtained from CT scans for the measurements of chord length along the 0° axis, pedicle diameter (perpendicular to pedicle axis), and vertebral body length. These can be seen to fall very close to the y = x line.

DISCUSSION

Improved detail of vertebral anatomy is presented here, relative to transpedicular placement of screws for use in surgical stabilization of the spine between T9 and L5.

The most important dimensions for pedicle fixation are those of pedicle diameter and chord length, because these dictate the upper limit to transpedicular screw diameter and length. Screw diameter is important because it limits screw strength. Both screw diameter and length are important because they influence screw-bone interface strength. Stated simply, the larger the screw diameter and length, the greater the surgical fixation strength. We have shown here that screws of 5 mm, or even 6-7 mm for the lower lumbar region, are compatible with anatomic constraints. Until now, screws of smaller diameter have been used,[6-9] and there has been some concern over screw breakage.[8]

Selection of the proper diameter of transpedicular screw is an important issue for safe surgical placement. Therefore, we were interested in whether the effective pedicle diameter was influenced very much by angle of screw placement. Figure 6 shows that no significant difference exists between any of the three measurement methods, for each vertebral level. Thus, the angle used for measurement of pedicle diameter is not particularly important. More important, after screw diameter selection, unintentional changes in

Table 2. Pedicle Diameter Distribution for Each Vertebral Level (Diameter Perpendicular to Pedicle Axis)

Level	*# of Pedicle*	*3.00-3.99 mm*		*4.00-4.99 mm*		*5.00-5.99 mm*		*6.00-6.99 mm*		*7.00-7.99 mm*		*8.00-8.99 mm*		*9.00-19.38 mm*	
T9	14	2	14%	1	7%	2	14%	3	21%	1	7%	2	14%	3	21%
T10	18	2	11			2	11	7	39			2	11	5	28
T11	22					3	14	4	18	3	14	9	41	3	14
T12	24					5	21	5	21	3	12	5	21	6	25
L1	22	2	9			4	18	4	18	3	14	6	27	3	14
L2	14									1	7	9	64	4	28
L3	24							2	8	3	12	4	17	15	62
L4	24											1	4	23	96
L5	20													20	100
Mean % (unweighted)			3.8		0.8		8.7		15		7.3		22.1		43.1

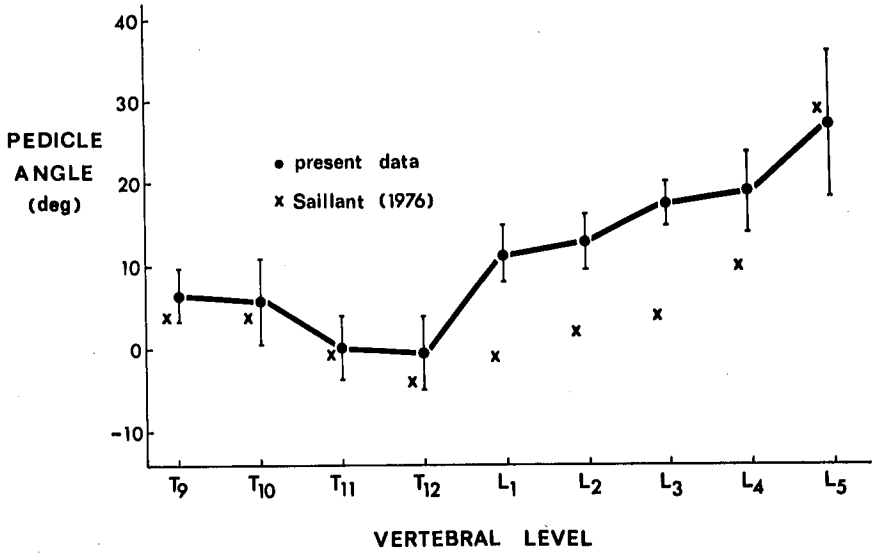


Fig 6. Pedicle axis angle relative to sagittal plane for each vertebral level. Means ± 1 SD are shown, compared with means ("x") from Saillant (1976).

screw orientation during screw placement are also not particularly important, insofar as diameter is concerned.

Measurements obtained preoperatively by CT scan on individual patients yield values that are quite accurate, both as demonstrated here (Figure 7) and as one would expect. If anything, CT measurements of pedicle diameter would tend to be a little smaller than actual because the medial and lateral pedicle cortices are convex in the frontal plane, and a typical slice thickness of 4 mm would average together bone tissue at the maximum diameter with bone tissue above and below this.

The dependence of chord length (and thus upper limit to screw length) upon angle relative to sagittal plane has been documented here for the first time and is of interest in terms of safe placement of transpedicular screws. Recommendations have been made by others for using the 0° orientation for screw placement.[6-8] However, the factual basis for this recommendation has not been reported. If a screw is placed more anterolaterally than intended, the tip of the screw may penetrate the anterior vertebral body cortex, risking damage to important adjacent structures. Shown here (Figure 2), however, is that even a 15° change anteromedially in placement angle results in only an 8-10 mm increase in length. We have reported elsewhere[3] that strength of fixation is not very sensitive to screw length (depth of penetration). Thus, during placement of a screw of chosen length, priority can be given to safety: erring slightly anteromedially will be safer and yet will still produce strength of fixation quite close to the maximum possible strength. Of course, excessive anteromedial angulation may produce interference between the tips of the right and left screws.

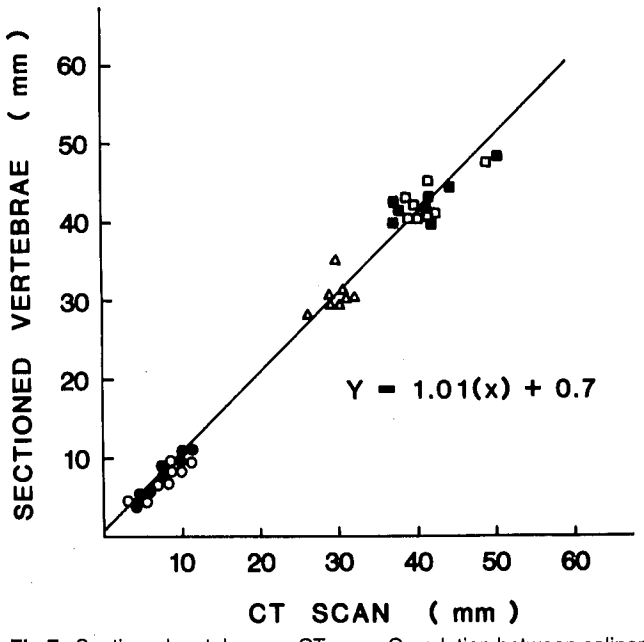


Fig 7. Sectioned vertebrae vs CT scan. Correlation between caliper measures and CT scan measurements on cadaver vertebrae.

Because the chord length is relatively constant from T9 to L5, the range of lengths of transpedicular screws needed is not large. This combined with the fact that fixation strength is not greatly dependent upon depth of penetration,[3] means that only a few different screw lengths are needed.

Pedicle angle (Figure 6) is shown to depart significantly from the sagittal plane in the lumbar region. This obliquity is of interest for two reasons. First, the pedicle diameters tend to be slightly larger for screw placement along the pedicle axis rather than along the 0° axis (straight-ahead position), at least in the lumbar region (Figure 5). Second, the right and left transpedicular screws at each vertebral level may be placed along this oblique axis, thus producing a "toenailing" effect. This substantially increases the resistance of the pair of screws against posterior pull-out, as well as resistance to lateral translation of one vertebra relative to a subjacent one.

Some differences exist between our pedicle angle mean values and those of Saillant[9] in the lumbar region, probably because pedicle angle has the least explicitly defined measurement method of all the measures presented here. Despite these differences, the trend with vertebral level is the same.

Pedicle length (Figure 3) is important because it defines the minimum length of screw needed to obtain a grip on the entire pedicle. The pedicle itself seems to be a stronger site for screw placement than the vertebral body. Even though both locations contain cancellous bone, the trabeculae in the pedicle appear to be stronger and thicker.[2] In addition, the pedicle cortex is quite thick, allowing the screw threads in many cases actually to engage cortical bone. The pedicle length ranges from 40-46% of the chord length, thus a transpedicular screw should be at least 46% of the chord length to obtain a grip along the full length of the pedicle.

Preoperative measurement of pedicle diameter can be accurately

performed from CT scans (Figure 7). Measurements of pedicle diameter from frontal plane (0° orientation) roentgenograms have not been reported here. Although the 0° pedicle diameters in Figure 7 are not significantly different from those measured at various obliquities (pedicle axis and 15°), they were obtained from CT scans and may bear little relation to values obtained from frontal plane roentgenograms. Proper screw length can either be measured directly on CT scan or can be estimated based on AP length of vertebral body measured on sagittal plane roentgenograms.

CONCLUSIONS

1. The growing interest in transpedicular screw fixation for spinal stabilizing implants is the basis for the improved morphometric details reported here.

2. Various, explicitly defined measures of interest have been obtained from clinical CT scans of human vertebrae from T9 through L5.

3. Pedicle diameters are large enough in most cases to allow substantially larger screw diameters than currently used and would improve strength of the transpedicular screw.

4. Chord length (posterior cortex of lamina to anterior cortex of body) is fairly constant between individuals and between vertebral levels. Thus, the range of screw lengths needed is not large.

5. The pedicle axis is oriented anteromedially, only slightly between T9 and L1; then it increases from L1 to L5.

6. Chord length along the 15° axis is approximately 1 cm longer than along the 0° axis. Screw placement direction should err, if at all, in the anteromedial direction, because this will decrease the risk of anterior cortex penetration.

7. The oblique orientation of the pedicle axis allows oblique screw placement, and thus an interlocking or "toenailing" effect between the right and left screws in each vertebra.

8. Preoperative determination of transpedicular screw length can be made, either by direct measurement from the patient's CT scan or by prediction from vertebral body anteroposterior diameter, as seen for example on sagittal plane roentgenogram.

REFERENCES

1. Dunn HK: Anterior stabilization of thoracolumbar injuries. Clin Orthop 89:116-124, 1984
2. Gallois M, Japiot P: Architecture interieure des vertebres. Rev Chir [Paris] 63:688-708, 1925
3. Krag MH, Beynnon BD, Pope MH, et al: An internal fixator for posterior application to short segments of the thoracic, lumbar, or lumbosacral spine. Clin Orthop 203:75-98, 1986
4. Magerl F: External skeletal fixation of the lower thoracic and the lumbar spine, in Current Concepts of External Fixation of Fractures. Edited by HK Uhthoff, E Stahl. New York, Springer-Verlag, 1982, pp 353-366
5. Magerl F: Clinical application on the thoracolumbar junction and the lumbar spine, in External Skeletal Fixation. Edited by DC Mears. Baltimore, Williams and Wilkins, 1983, pp 553-575
6. Roy-Camille R, Saillant G, Berteaux D, Marie-Anne S: Early management of spinal injuries, in Recent Advances in Orthopedics, vol 3. Edited by B McKibbin. New York, Churchill Livingstone, 1979, pp 57-87
7. Roy-Camille R, Saillant G, Berteaux D, Salgado V: Osteosynthesis of thoraco-lumbar spine fractures with metal plates screwed through the vertebral pedicles. Reconstr Surg Traumatol 15:2-16, 1976
8. Roy-Camille R, Saillant G, Marie-Anne S, Mamoudy P: Behandlung von Wirbelfrakturen und-luxation am Thorako-lumbalen Ubergang. Orthopade 9:63-68, 1980
9. Saillant G: Etude anatomique des pedicles vertebreau, application chirurgicales. Rev Chir Orthop Traumatatol 62:151-160, 1976
10. Zindrick MR, Wiltse LL, Doornik A, et al: Analysis of the morphometric characteristics of the lumbar and thoracic pedicles. Spine 2:160-166, 1987

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